

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Midterm
Duration: 1 hour 20 min

Semester: Spring 2025
Full Marks: 40

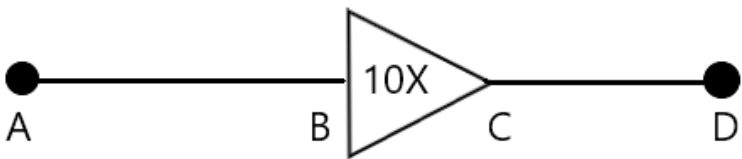
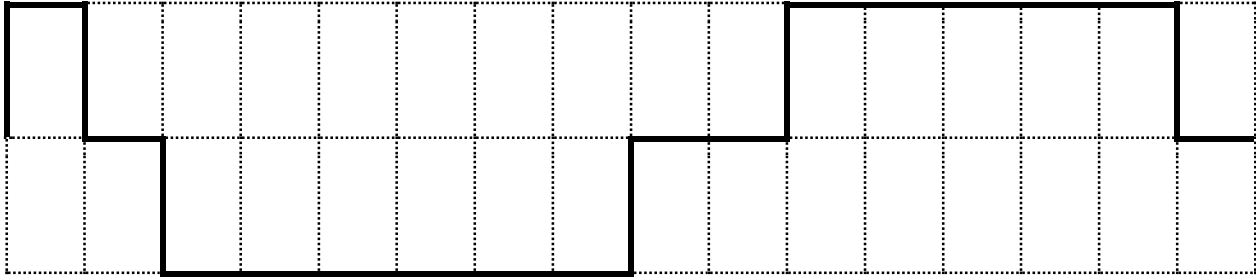
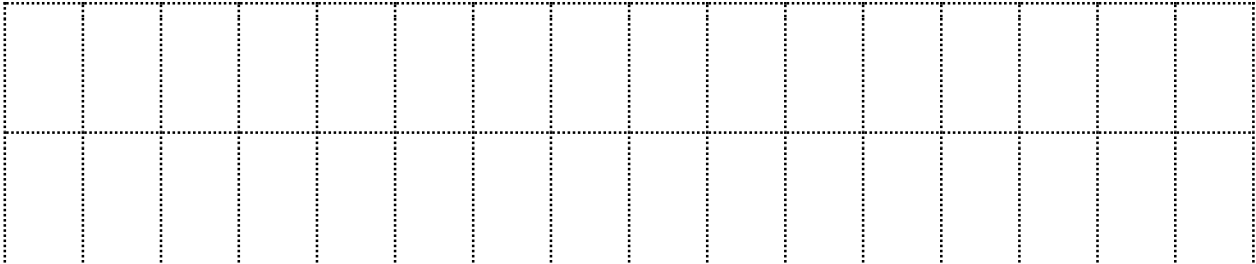
CSE 320: Data Communications

Answer the following questions [Total 3].
Figures in the right margin indicate marks.

SET B

Name:	ID:	Section:
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1.	a)	[CO1] Design a network infrastructure for a Software Development Company that consists of three main departments: i) Development Department – This department includes four teams: Frontend Team, Backend Team, QA Team and DevOps Team. Each team is directly connected to other teams, ensuring that if one connection fails, communication can still continue through alternate paths. ii) Database & Cloud Infrastructure Department – This department consists of three subunits: Primary Database Server, Backup Server and Cloud Storage Nodes. To ensure high availability and fault tolerance, all servers are connected through a centralized system, ensuring efficient data flow and resource management. iii) Administration & Support Department – This department includes HR & Finance, IT Support, Meeting Room and Project Management Office. Here, data flows through a common communication channel where data is transmitted sequentially and each device receives the information before passing it along to the next. However, if one device fails, it might disrupt communication unless a backup mechanism is in place. To connect all three departments, the company uses a single backbone connection, allowing data to travel efficiently between them. Identify the networking structure used in each department based on the characteristics described above. Draw a hybrid network topology diagram representing this setup, labeling all units and subunits accordingly. Calculate the total number of connection links.	[6]
	b)	[CO1] Highlight the key functionalities of the Data link layer and Transport layer of OSI network model. Mention which addresses are used here and why using both is necessary?	[4]

2.	a)	 [CO2] Suppose, a signal is transmitted from source A to destination D. One 10 times amplifier is placed between B and C. Attenuation along path A to B is 6.99 dB, and along path C to D is 3.98 dB. If power at destination D is measured to be 8W, find out the power of the signal at source A.	[5]
	b)	[CO2] Suppose, while playing Call of Duty, you are downloading a frame of size 1MB (mega-byte) from a server that is 10,000 km away. The fiber-optic transmission medium from the server to your house contains 10 routers, each of which has a queuing time of 1 ms and a processing time of 2 ms. The speed of light inside fiber optic cable is 2×10^8 m/s. If the latency (total delay) is found to be 4.08 s, calculate the throughput of your network.	[5]
	c)	[CO2] Consider a transmission channel in which the value of SNR is infinity. What can be deduced from this statement? Suppose, you want to download a folder from your google drive that contains 100 images. Each image has an average size of 200 kB (kilo-byte). However, your network has a noisy channel with an SNR of 63. Consider the bandwidth to be 2 MHz. Calculate the maximum capacity bitrate of your network.	[5]
3.	a)	[CO2] Sami and his friend Shafi come across a three-level signal and both attempt to decode it. Sami claims that the signal follows the Bipolar AMI encoding scheme, while Shafi argues that it is encoded using MLT-3. Determine which of them is correct and then decode the given signal. [answer in question paper] 	[4]
	b)	[CO2] Deduce the appropriate scrambling method for the signal given in 3. a) and redraw the signal using it. [answer in question paper] 	[4]

c) **[CO2] Design** a 3B/4B Block Coding scheme while following the **conditions below**. For convenience, the 4B encoded sequence for some 3B data sequences are already given. [Any valid combination following the conditions are correct] [7]

- The encoded sequence cannot contain more than two 0
- The encoded sequence cannot contain three or more consecutive 1

Data Sequence	Encoded Sequence
000	1101
001	
010	1011
011	
100	
101	0011
110	
111	

Based on the above block coding scheme, encode the following dataword into its corresponding codeword.

110010111001100

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